

Week 12: Global State Management

1. Context API (*createContext, useContext*)

The Context API is used for sharing data across multiple components without passing props manually. `createContext` creates a shared data store, while `useContext` allows components to access that data. This makes state handling simpler in medium-sized applications.

2. Introduction to Redux (*Store, Actions, Reducers*)

Redux is a state management library used for handling application-wide data. The store is the central place where state is stored. Actions define what changes should happen, and reducers update the state based on those actions. Redux helps manage predictable and scalable application data flow.

3. Redux Toolkit

Redux Toolkit is the modern and recommended way to write Redux code. It reduces boilerplate code and simplifies store setup, reducers, and action creation. Developers can manage state faster and more efficiently using Redux Toolkit.

4. Middleware (*redux-thunk*)

Middleware in Redux is used to handle asynchronous tasks like API calls. `redux-thunk` allows writing functions inside actions to perform async operations before updating the state. This improves flexibility and makes state updates more manageable.

Conclusion

This week focused on global state management. By learning Context API, Redux, Redux Toolkit, and middleware like `redux-thunk`, better control over application-wide state handling was achieved.